

Moodle Administration Basic

Modules 1, 2, 3, 5, and 6 Foundation training for Moodle administrators

Training roadmap

- Module 1: Introduction to Moodle
- Module 2: Moodle Administration Basics
- Module 3: Understanding Moodle Technical Components
- Module 5: Customization and Extensibility
- Module 6: Maintenance and Troubleshooting

Learning outcomes

- Explain Moodle as a learning management system and web application
- Navigate core administration areas and support common user/course tasks
- Identify Moodle code, database, moodledata, cron, and service dependencies
- Apply safe customization and plugin evaluation practices
- Use structured maintenance and troubleshooting workflows

Module 1: Introduction to Moodle

Purpose: understand what Moodle is, who uses it, and how the main parts fit together.

What is Moodle?

- Open-source learning management system for delivering and managing learning
- Supports courses, users, activities, resources, grades, and communication
- Used by universities, schools, training providers, and organizations
- Administrators support both the learning platform and its technical environment

Moodle user roles at a glance

- Students access content, submit work, join activities, and view grades
- Teachers create course content, manage activities, and assess learners
- Managers support categories, programs, departments, or delegated operations
- Site administrators configure the platform and coordinate technical support

High-level Moodle flow

- User opens Moodle in a browser
- Web server runs Moodle application code
- Moodle checks authentication, enrolment, roles, permissions, and context
- Database stores records and settings; moodledata stores uploaded/generated files
- Cron processes background jobs such as notifications and scheduled tasks

Module 1 demo ideas

Show front page and dashboard Open a sample course: sections, activities, participants, grades Briefly show Site administration without changing settings
Draw a simple browser → web server → Moodle → database/moodledata diagram

Module 1 key takeaways

Moodle is both an LMS and a web application Different roles experience Moodle differently Core platform dependencies include code, database, moodledata, cron, and infrastructure Administrators need application awareness and basic infrastructure awareness

Module 2: Moodle Administration Basics

Purpose: perform common administrative tasks safely and understand access concepts.

Site administration orientation

- Main configuration area for Moodle
- Use search and categories to find settings
- Do not try to memorize every setting at first
- Document meaningful changes, especially on production sites

User management basics

- Create, search, suspend, and update user accounts
- Distinguish authentication from enrolment
- Review profiles carefully before changing accounts
- Use test accounts in training or staging environments

Roles, permissions, and context

- A role is a collection of permissions
- Context determines where a role applies: site, category, course, activity, or user
- Avoid overusing site administrator access
- Assign the minimum role needed for the task

Course management basics

- Courses are organized into categories
- Visibility, enrolment, and roles affect whether users can access a course
- Courses contain resources, activities, participants, grades, and reports
- Common resources/activities include File, Page, Forum, Assignment, Quiz, and URL

Module 2 guided walkthrough

Open Site administration Search for a user Create or inspect a course category and course Enrol a user as a student Assign a teacher role in a course Turn editing on and add one resource and one activity

Module 2 troubleshooting lens

Login problem: can the account authenticate? Course access problem: is the course visible and is the user enrolled? Permission problem: does the user have the correct role in the correct context? Change control: what changed, who changed it, and when?

Module 3: Understanding Moodle Technical Components

Purpose: know the platform components that must be protected, monitored, and backed up.

Main technical components

- Moodle application code directory
- config.php with environment-specific settings
- Database for structured records and configuration
- moodledata for uploaded/generated files and cache areas
- Cron and scheduled tasks for background processing
- Web server, PHP, email service, authentication providers, and integrations

Moodle code directory

- Contains core Moodle files and installed plugin code
- Should be managed consistently during upgrades
- Incorrect file changes can break plugins or themes
- Do not expose credentials while demonstrating config.php

Database overview

- Stores users, courses, enrolments, grades, activity data, settings, logs, and metadata
- Database availability directly affects the whole site
- Backups must include the database, not only the application files
- Avoid manual database changes unless approved and understood

moodledata overview

- Stores uploaded files and generated files outside the web-accessible code directory
- Disk space problems can break uploads, backups, and cache generation
- Permissions and ownership must allow Moodle to read and write safely
- Backup strategy must include moodledata

Cron and background jobs

- Runs scheduled and ad hoc tasks
- Supports notifications, cleanup, completion, analytics, and other background processing
- Delayed or broken cron can cause email, forum, and scheduled-task issues
- Administrators should know where scheduled task status is displayed

Backend workflow example

- Student requests a course file
- Moodle validates login, enrolment, role, permission, and activity/course visibility
- Database provides records and file metadata
- File content is served from moodledata
- Logs may record the action for later review

Module 5: Customization and Extensibility

Purpose: customize Moodle while managing risk, compatibility, and maintainability.

Customization options

- Themes for appearance and user experience
- Theme settings for logo, colors, menus, and branding
- Plugins for new activities, blocks, reports, authentication methods, enrolment methods, and more
- Integrations for SSO, email, storage, video, proctoring, payment, reporting, and external systems

Plugin evaluation checklist

- Is it compatible with the current and target Moodle versions?
- Is it actively maintained and widely used?
- Does it meet security, privacy, and institutional requirements?
- Can it be tested in staging before production?
- What is the rollback and upgrade plan?

Safe customization practices

- Prefer configuration and supported theme settings before custom code
- Test changes in staging first
- Document the business reason, owner, settings, files, and dependencies
- Avoid unknown plugins in live demonstrations
- Plan customizations as part of the upgrade lifecycle

Module 6: Maintenance and Troubleshooting

Purpose: keep Moodle healthy and solve incidents with evidence instead of guesses.

Routine maintenance checklist

- Confirm backups are successful and restorable
- Review cron and scheduled task status
- Monitor disk space, server resources, database health, email delivery, authentication, and integrations
- Apply updates using approved change windows
- Review logs and recurring incidents

Common issue categories

- Login and authentication issues
- Course access, enrolment, visibility, and role issues
- Upload, storage, backup, or disk-space issues
- Performance, cache, database, or server-resource issues
- Email, notification, cron, and integration issues

Basic troubleshooting workflow

- Define the symptom and scope
- Identify affected users, courses, activities, time period, and recent changes
- Check logs, settings, enrolments, roles, cron, and system status
- Make one controlled change at a time
- Document findings, resolution, and prevention steps
- Escalate when production risk, data integrity, security, or infrastructure expertise is involved

Capstone scenario discussion

- A teacher says students cannot access a course. What should you check first?
- A scheduled forum announcement was not emailed. Which components are involved?
- A course backup fails. What evidence would you collect?
- A new plugin is requested. What questions must be answered before installation?

Final administrator habits

- Use staging for risky changes
- Back up code, database, and moodledata
- Prefer least privilege for roles
- Monitor before users report incidents
- Document changes and incidents
- Escalate with clear evidence

Thank you

- Questions and discussion